Knapsack Sprayer



Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **High** 8 labs

Covers these lab items: Knapsack Sprayer

What it is used for

A knapsack sprayer applies approved liquid treatments to crops, plants, or other designated areas. Correct mixing, pressure control, protective equipment, and cleaning help prevent exposure, leaks, and environmental contamination.

Main parts

- Tank stores the approved spray liquid.
- Filler opening and cap allow safe filling and closure.
- Pump or pressure mechanism builds the required spray pressure.
- Lever or motor control operates the pump on applicable models.
- Hose carries liquid from the tank to the spray lance.
- Lance and nozzle direct the spray onto the target.
- Shoulder straps support the tank during carrying.

Before you start

1. Read the product label and sprayer manual before preparing any spray mixture.
2. Use only approved substances and follow the specified dilution and application instructions.
3. Wear the required PPE, including chemical-resistant gloves, protective clothing, footwear, and eye protection.
4. Inspect the tank, cap, straps, hose, seals, lance, nozzle, and pump for damage or leaks.
5. Prepare the mixture in a ventilated area away from food, drinking water, children, and animals.
6. Check wind direction and avoid spraying in strong wind or toward people and sensitive areas.

How to use

1. Confirm the sprayer is clean and suitable for the intended product.
2. Prepare the spray mixture according to the product label and fill the tank without overfilling.
3. Close the cap securely and check for leaks before carrying the sprayer.
4. Adjust the shoulder straps so the tank sits securely without restricting movement.
5. Build pressure using the approved pump or start the motor according to the manual.
6. Test the spray pattern in a safe designated area before application.
7. Hold the lance toward the target and apply the recommended amount evenly.
8. Keep the nozzle away from your body and avoid breathing spray mist.
9. Stop spraying if leaks develop, the nozzle blocks, or wind changes create exposure risk.
10. Release pressure safely, empty and clean the sprayer as directed, and wash hands after removing PPE.

⚠ Safety

- Never spray without the PPE specified on the product label.
- Do not mix chemicals or use unlabelled substances in the tank.
- Never point the nozzle at people, animals, food, or drinking-water sources.
- Do not eat, drink, or smoke while mixing or applying chemicals.
- Release pressure before opening the tank, removing the nozzle, or clearing a blockage.
- Keep children and animals away from the treated area for the period stated on the product label.

Care & storage

- Rinse and clean the tank, hose, lance, and nozzle according to the product label and manual.
- Use a suitable brush or cleaning method for blocked nozzles; never blow through them by mouth.
- Inspect seals, straps, valves, and hoses regularly and replace damaged parts.
- Store the clean, dry sprayer securely and dispose of leftover mixture and rinse water as directed.

Common mistakes to avoid

- Mixing chemicals without reading the label or measuring the required dilution.
- Opening a pressurised tank or clearing a nozzle with the mouth.
- Spraying against the wind or without the required protective equipment.
- Disposing of leftover chemicals or rinse water into drains or water bodies.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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